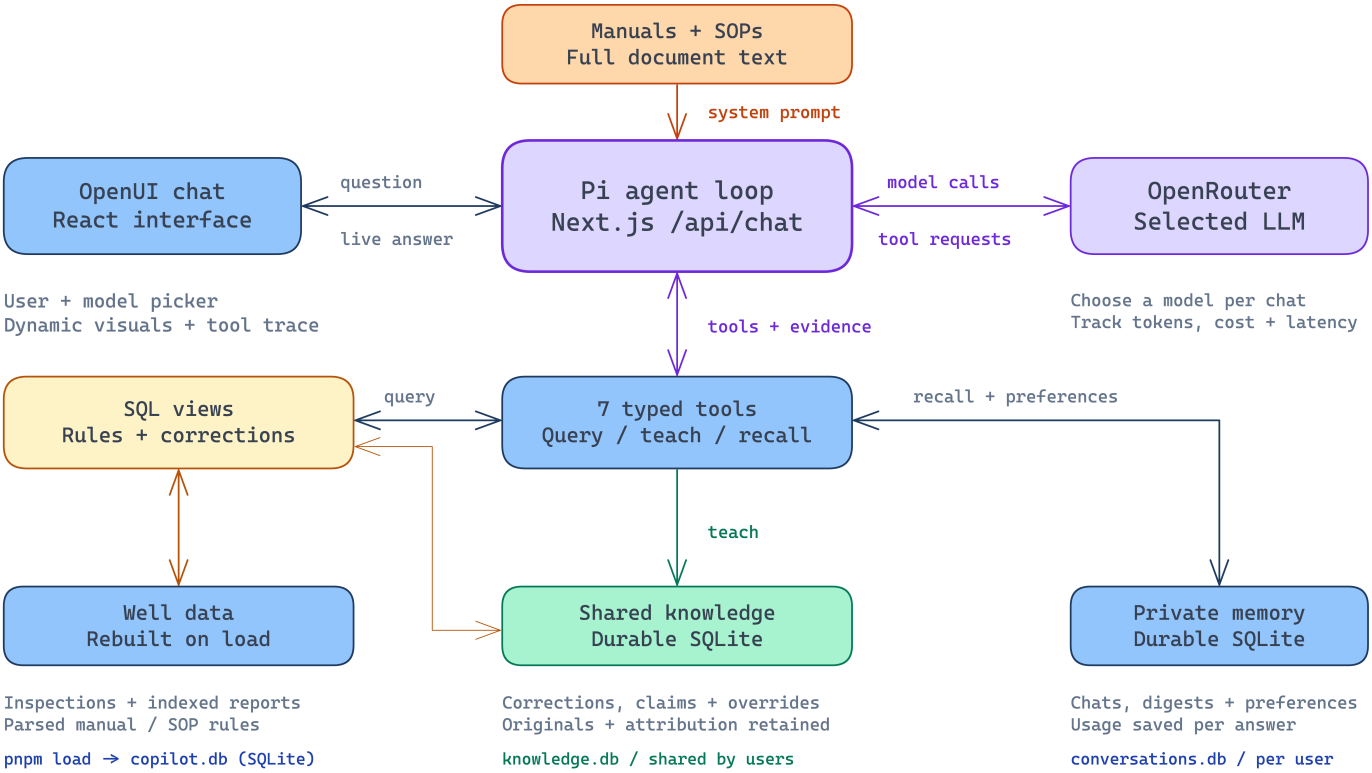
Asset Intelligence Copilot

Ask about wells. Get cited answers, useful visuals and lasting memory.



Pi + runtime tools. One Pi agent per user/conversation runs the model-tool-result loop at `/api/chat`. Five fleet tools handle snapshots, read-only SQL, report search/read and shared knowledge; two user-bound tools manage recall and preferences. Pi executes model-chosen calls and arguments, streaming results into the visible tool trace.

Dynamic visuals. OpenUI renders cited cards, tables, charts, maps and heatmaps. For custom requests, the model generates a Vega-Lite `Plot`: threshold lines, highlighted readings, timelines or layered charts. Follow-ups reshape the previous visual using the same data; map styles and palettes are configurable.

Manuals and SOPs: two paths. Full document text enters the system prompt with the glossary, view schema and user context. Parsed rules also drive SQLite views for production, downtime, spill classes and compliance. SQL calculates; answers cite the governing document section.

Evidence storage. Three SQLite files separate rebuildable fleet data, shared knowledge and private conversations. FTS5 searches repair reports; tools return snippets, full text and extracted containment/compliance facts alongside structured records and operational notes.

Shared learning and conflicts. Corrections retain original values, replacements, author and reason. Accepted corrections and well-type overrides update everyone's derived results at their exact scope. Contradicted claims remain unconfirmed; disputed notes preserve both accounts. Knowledge survives dataset reloads.

Private, durable memory. Pi holds live state in RAM; SQLite saves transcripts, answer digests and usage after each turn. New chats receive up to 15 of that user's recent digests. `recall_my_conversations` searches older digests or reads original answers for ordered selections. Missing matches require clarification. Saved transcripts resume after restarts.

Personal preferences. `my_preferences` saves, lists, replaces or removes private presentation rules across chats, such as "always use red for wellhead temperature". A one-off request takes precedence without changing standing preferences or shared well facts.

Models, cost and new data. OpenRouter offers five models, defaulting to GPT-6 Luna; conversations keep their model. Per-answer tokens, cost and latency cover every model call. Dataset roots merge by ID, later roots win, and missing required manual rules fail the load explicitly.

Next improvements. Compact long active transcripts, retrieve manual sections for larger corpora, and add authenticated identity beyond the demo user picker.

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Final questions: results and limitations

GPT-6 Luna answered all 11 numbered questions (21 turns) on the 600-well test dataset, as of 15 January 2027, preserving the prescribed user/session sequence. These are actual agent responses, not corrected reference answers. No grader or scores were used for this capture. Completed turns report $0.36 in model usage; this excludes interrupted requests and is not a billing reconciliation.

Question	Observed answer and limitation
Q1. Reservoir pressure	Declined to invent a pressure map because the records contain temperatures, not reservoir pressure.
Q2. Best producer	Distinguished missing oil volumes from an inspection-count proxy; the proxy cannot identify a unique top producer.
Q3. Gas reports	Produced dated response/deadline evidence; completeness and first-response timestamp matching remain weak.
Q4. Bypasses	Joined notes and approval memos into a status table; historical intervals and extensions need deterministic validation.
Q5. Regulator letter	Cited sampling evidence, but some compliance reasons contradict their quoted reports.
Q6. Status conflicts	Compared inspection and operator-note evidence; imported correction handling and precedence are not consistently reliable.
Q7. Maintenance priorities	Quoted warning notes and ranked candidates, then selected three; some listed wells already have repairs, weakening the required filter.
Q8. Attribution and recall	Exercised shared replacement reports and private preferences; the pump inventory contains duplicates and replaced pumps, with inconsistent attribution.
Q9. Conversation continuity	Returned to earlier subjects and declined to invent an H2S reading; team-lead lookup and cross-topic evidence remain incomplete.
Q10. Delay reasons	Explicitly limited the answer to report-verified examples and declined a misleading fleet-wide chart; complete aggregation is unsupported.
Q11. Regional heat	Compared raw temperatures and ambient conditions; a complete within-well-type, month-by-month comparison is missing.

What worked. Evidence-bearing tables, durable user/session continuity, provenance and honest refusals for missing measurements. All full responses are retained for inspection, including their errors.

What did not, and why. Views do not fully handle historical well-type changes or dataset-labelled corrections. Relevance-ranked report search is not an exhaustive aggregation interface. Long model investigations still omit evidence, mishandle inventories and fail some requested visuals. Stronger instructions did not remove these data/tool limits.

Next changes. Apply type history and imported corrections in shared data logic; extract installed-pump history and delay reasons into queryable fields; add deterministic checks for status precedence, time windows and list completeness. Keep presentation preferences private and operational claims attributed.